

Subject Description Form

Subject Code	EEE2001
Subject Title	Applied Electromagnetics
Credit Value	2
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To introduce students to the physical laws governing the electromagnetic phenomena commonly encountered in electrical and electronic engineering systems. 2. To familiarize students with the techniques for solving problems in electromagnetics. 3. To provide students with the fundamental understanding of electromagnetic field theory required for pursuing the EEE programme.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand that electromagnetism is based on Maxwell's equations, interpret the physical meaning and phenomena behind Maxwell's equations, and know the meanings and basic relationships of physical quantities in electromagnetism. b. Analyse electromagnetic phenomena related to electrical and electronic engineering systems by selecting the most appropriate laws/theorems/ techniques. c. Have hands-on experience in electromagnetic measurements.
Subject Synopsis/ Indicative Syllabus	1. Static fields: Electrostatics: Electric fields, Coulomb's law, Gauss's law, potential, capacitance and energy storage. Magnetostatics: Biot-Savart law, magnetic fields, Ampere's circuital law. Force on a current-carrying conductor, Lorentz force. 2. Time-varying fields: Faraday's Law and Lenz's Law; self-inductance, mutual inductance and stored energy. 3. Mathematical preliminaries: Vectors analysis and coordinate systems. The operators grad, div and curl. Concept of line, surface and volume integrals. 4. Maxwell's equations and EM waves: Maxwell's equations in integral form as a restatement of fundamentals. Differential form. The continuity equation. The displacement current. 5. Material media: Dielectric material: dipole, polarisation, permittivity and capacitors. Ferromagnetism: magnetisation curve, permeability, hysteresis and saturation. Boundary conditions. Magnetic circuits: magneto-motive force, reluctance and permeance. Laboratory Experiments: Field plotting using the Electrolytic tank. Field plotting using the resistive paper.

Teaching/Learning Methodology	Lectures and tutorials are the primary means of conveying the basic concepts and theories. Experiences on analysis and practical applications are gained through experiments and using software, in which the students are expected to solve problems with critical and analytical thinking. Experiments are designed to supplement the lecturing materials so that the students are encouraged to take extra readings and to look for relevant information. Software is used to help the students to understand the physical meanings of mathematical equations.					
	Teaching/Learning Methodology		Outcomes			
			a	b	c	
	Lectures		✓	✓		
	Tutorials		✓	✓		
	Experiments		✓	✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% weighting	Intended subject learning outcomes to be assessed		
				a	b	c
	1. Examination		60%	✓	✓	
	2. Class Test		18%	✓	✓	
	3. Assignment		12%	✓	✓	
	4. Laboratory performance & report		10%	✓	✓	✓
	Total		100%			
	It is a fundamental subject of electromagnetics. The outcomes on physical concepts and analysis are assessed by the usual means of examination, assignment and test whilst those on analytical skills and problem-solving techniques, as well as technical reporting and teamwork, are evaluated by experiments, software application and the reports.					
Student Study Effort Expected	Class contact:					
	▪ Lecture/Tutorial			22 Hrs.		
	▪ Laboratory			4 Hrs.		
	Other student study effort:					
	▪ Laboratory preparation/report			6 Hrs.		
	▪ Self-study			38 Hrs.		
	Total student study effort			70 Hrs.		
Reading List and References	Reference books: 1. W. H. Hayt and J. A. Buck, Engineering Electromagnetics, 8 th Edition, Boston: McGraw Hill, 2012.					

	2. N. N. Rao, Elements of Engineering Electromagnetics, 6th Edition, Pearson Education International, 2006. 3. F. T. Ulaby and U. Ravaioli, Fundamentals of Applied Electromagnetics, 7th Edition, Pearson Education International, 2015. 4. K. E. Lonngren, etc., Fundamentals of Electromagnetics with Matlab, 2nd Edition, Scitech Publishing, Inc., 2007.
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